PROPER LIMIT OF PROPER FINITE PRESENTATION MORPHISMS (NOETHERIAN INDEX CASE)

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Editorial note. Author omission kept verbatim: 'These morphisms form the transition morphisms of the system. Some details omitted.' The construction steps are given; the compatibility verification of the transitions is unstated.

Allowed standard prerequisites (editorial).

- Descent of finite presentation. Limits, Lemma lemma-descend-finite-presentation (same chapter)

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a proper morphism with S quasi-compact and quasi-separated. Then there exists a directed set I, an inverse system $(f_i : X_i \to S_i)$ of morphisms of schemes over I, such that the transition morphisms $X_i \to X_{i'}$ and $S_i \to S_{i'}$ are affine, such that f_i is proper, such that S_i is of finite type over $\mathbf{Z}$, and such that $(X \to S) = \lim(X_i \to S_i)$.*

Proof. By Lemma 09ZR we can write $X = \lim_{k \in K} X_k$ with $X_k \to S$ proper and of finite presentation. Next, by absolute Noetherian approximation (Proposition 01ZA) we can write $S = \lim_{j \in J} S_j$ with S_j of finite type over $\mathbf{Z}$. For each k there exists a j and a morphism $X_{k,j} \to S_j$ of finite presentation with $X_k \cong S \times_{S_j} X_{k,j}$ as schemes over S, see Lemma 01ZM. After increasing j we may assume $X_{k,j} \to S_j$ is proper, see Lemma 081F. The set I will be consist of these pairs (k, j) and the corresponding morphism is $X_{k,j} \to S_j$. For every $k' \geq k$ we can find a $j' \geq j$ and a morphism $X_{j',k'} \to X_{j,k}$ over $S_{j'} \to S_j$ whose base change to S gives the morphism $X_{k'} \to X_k$ (follows again from Lemma 01ZM). These morphisms form the transition morphisms of the system. Some details omitted. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.